

Children's ability concepts: Their development, content, and consequences

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Abstract

In this chapter, we summarize theoretical and empirical work on the development of ability concepts in children. We first examine the form of children's basic concept of ability, asking whether it undergoes major differentiation during development or whether, instead, a near adult-like ability concept is available early on. We then ask when in development children's ability beliefs begin to exhibit coherence and motivational force. The body of evidence reviewed here points to previously unnoticed sophistication, depth, and coherence in young children's ability concepts. We close with a discussion of how theoretical and empirical attention to early ability concepts might advance our understanding of motivational processes in early childhood, as well as offer insights into how to optimize these processes.

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Ability is central to who we are and what we do. Even in the course of an ordinary day, we may think about how much of it we have, how much of it others have, and what this might mean for our, and others', ability to master skills and succeed on important tasks. Whatever ability may actually be, or however it may contribute to success, as a concept it plays a central role in everyday thought and behavior. Given its centrality, it is critical to examine both how we come to understand ability and the potential implications of these understandings.

An examination of this sort is all the more pressing given that ability concepts figure prominently in nearly every contemporary psychological account of motivation (Bandura, 1982; Covington & Beery, 1976; Dweck, 1986; Weiner, 1985; Wigfield & Eccles, 2000). The extent to which we embrace challenges, persist after failure, and more generally try to achieve mastery, for instance, is intimately tied to our basic assumptions about what ability is like and what it does (Bian et al., 2018; Blackwell et al., 2007; Elliott & Dweck, 1988; Limeri et al., 2023; Rege et al., 2021; Yeager et al., 2019). In short, our concepts of ability are foundational to our motivation, to our decisions about our learning, and, ultimately, to what we achieve.

In light of their theoretical and practical significance, our goal in this chapter is to synthesize and evaluate evidence on early concepts of ability, focusing roughly on the grade school age range (5–11 years). Throughout, we consider classic and contemporary findings with the goal of arriving at a clear, comprehensive, and up-to-date overview of young children's thinking about ability. Such an endeavor, in our view, is needed to ensure that future research efforts are grounded in what is currently known about children's ability concepts. As we have foreshadowed, one question that readers might keep in mind in reading this chapter is: How similar is young children's ability-related thinking to that of older children's and adults'? To anticipate our conclusions, early research often pointed to dramatic differences, but newer methods have revealed fascinating similarities.

We discuss early concepts of ability in two major parts. We start by considering whether children have well-formed, adult-like ability concepts early on, or whether, instead, these concepts must be gradually acquired over early and middle childhood. After addressing the question of whether young children's ability concepts resemble adults', we then ask whether children's beliefs about ability are rudimentary and disconnected from their motivation and learning, or whether, instead, these beliefs are coherent and linked with achievement behaviors, such as challenge-seeking, early on. Against this backdrop, we conclude the chapter by outlining avenues for future research,

highlighting open questions whose answers can yield insight into how best to sustain children's motivation.

In addressing these questions, we find it useful to regard children's ability concepts as embedded in domain-specific, theory-like knowledge structures called "intuitive theories" (e.g., [Gopnik & Wellman, 2012](#); [Kinlaw & Kurtz-Costes, 2003](#); [Murphy & Medin, 1985](#); [Wellman & Gelman, 1992](#)). Like a scientific theory, an intuitive theory is used to interpret, explain, and predict everyday events within its scope ([Keil, 2024](#)). Here, it makes sense to assume that both children and adults have informal understandings of performance and the factors that contribute to it. This informal understanding can be thought of as an intuitive theory, comprising interconnected concepts such as ability, which can be used to explain what happened in the past (e.g., why did she fail the bar exam?) and make predictions about what might happen in the future (e.g., will she fail the bar exam again?). Importantly, with this perspective, and in keeping with cognitive development traditions, we can characterize developmental change, if any, in intuitive theories by asking how similar (vs. different) young children's ability concepts are relative to those of older children and adults (for a similar strategy, see [Carey, 1985, 2000](#)).



1. The development of children's ability concepts: Change or continuity?

Adults have intuitive understandings of ability as a cause, distinct from effort, that increases the likelihood of (but does not guarantee) good performance.¹ That is, adults' intuitive theory of performance has embedded in it *differentiated* concepts of ability and effort. When do children's intuitive theories begin to resemble this structure? Are the concepts embedded in this intuitive theory kept separate by children, or might they be conflated initially instead? These questions have implications not only for our basic understanding of conceptual development, but also for approaches to studying motivational processes in early childhood. If young children have an undifferentiated ability concept that includes effort, this might suggest that their motivation and learning behaviors (e.g., their responses to failure) look quite different from older children's and adults'. If, however,

¹ For our purposes here, we define ability as a personal resource that (a) increases the likelihood of, but does not guarantee, good performance and (b) is fairly consistent in its effects on performance across situations. One thing that distinguishes ability from other personal resources (e.g., effort) is that it is not under one's control at the time of performance. One's ability could change over longer temporal spans, but one cannot instantaneously increase their ability at will.

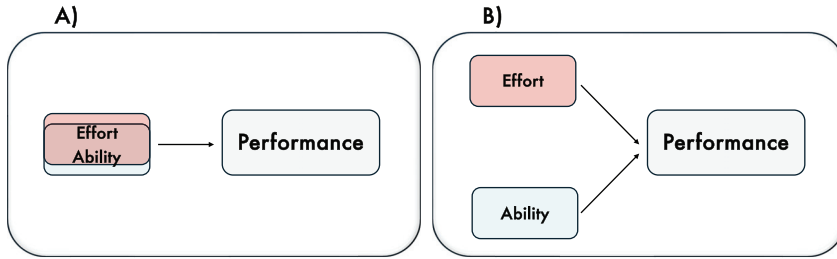


Fig. 1 Schematic of an intuitive theory of performance with ability and effort (a) as undifferentiated concepts versus (b) as differentiated concepts.

young children do have a distinct ability concept, this would suggest the early presence of a foundational aspect of motivation, and would motivate empirical examinations of how this concept is involved in motivational processes.

An influential theory put forth the idea that children's ability concepts develop quite gradually in a stage-like fashion, with early stages reflecting immature or misguided understandings and later stages reflecting mature, adult-like understandings. In particular, this canonical perspective suggested that children do not understand ability as a distinct source of performance until they reach 10 years of age (Nicholls, 1978; Nicholls & Miller, 1984a; 1984b). On this view, children start out with an undifferentiated concept of ability that is combined with something different: effort (see Fig. 1). The differentiated, mature concept of ability was thought to emerge only after a stage-like restructuring of children's intuitive theories (Nicholls & Miller, 1984a).

Accounts of this sort were not limited to children's understanding of ability. The idea that children's (initially jumbled) concepts become increasingly differentiated with age has been central to developmental psychology, and to cognitive development traditions especially (e.g., Marsh et al., 1984; Smith et al., 1985; Werner, 1957). This characterization has been applied to diverse sets of concepts: for example, to children's concepts of weight and size (Piaget & Inhelder, 1974), to their concepts of inanimate and animate, as well as inanimate and dead (Carey, 1985, 2000; Piaget, 1929), and to their concepts of numerosity and spatial arrangement (Piaget & Inhelder, 1969), to name a few. To foreshadow our conclusions, some of these characterizations were later challenged with evidence showing that young children's knowledge is often masked by (overlooked) features of the very procedures used to assess it (e.g., Gelman, 1990; Keil, 1979; McGarrigle & Donaldson, 1974; Nguyen & Gelman, 2002; Rose & Blank, 1974; Smith et al., 1985).

Evidence for the idea that children initially conflate ability and effort came mainly from a series of studies in which children were asked to make judgments about two students' abilities after being given information about how much effort they put into their work and about their performance outcomes (Chapman & Skinner, 1989; Miller, 1985; Nicholls, 1978; Nicholls & Miller, 1984b). In these studies, participating children were shown films of two students working side by side on schoolwork: One student is shown to work the entire time, while the other spends only some of the time working. It is eventually revealed that the students earned identical (and high) grades on their schoolwork, and at the film's conclusion, participating children are asked to respond to questions about the students' effort expenditure, ability, and so on.

Critically, when participants were asked to make judgments about the students' abilities, children younger than 10 usually said that the student who worked *harder* had more ability than the student who did not put in as much effort. The interpretation was that young children chose the student who put in more effort because effort and ability are not yet differentiated concepts: People who try harder are smarter. This account suggested that around age 10 is when these two concepts become neatly separated.

If young children really do think that a student who has to expend *more* effort for the same outcome has more ability, then their ability concepts are quite different from those of older children's and adults', who typically view the less-hardworking (but still high-achieving) student as having more ability. Further, this canonical finding was emerging alongside and fit well with results painting young children as characteristically lacking in sensitivity to ability-relevant information. For example, across a number of studies using a variety of tasks, young children's assessments of their own abilities and future performance were often notably and unreasonably high (Parsons & Ruble, 1977; Stipek & Tannatt, 1984; Yussen & Levy, 1975; see also Leonard & Sommerville, 2025; Xia et al., 2022). Claims from the canonical account could be used to explain this: If ability and effort are one conceptual unit, then securing a good performance outcome in the future is just a matter of trying harder.

Young children also seemed curiously resilient to failure—that is, failure did not seem to prompt the decrements in performance or persistence (Miller, 1985; Rholes et al., 1980) or the tempered expectancies for future success (Parsons & Ruble, 1977; Stipek & Hoffman, 1980) seen among older children. These findings were also interpreted against the backdrop of the canonical theory that children's concept of ability is not yet differentiated

from effort: Without a distinct concept of ability, failure or otherwise subpar performance is not an indictment of one's abilities and therefore cannot amount to a motivation-thwarting event.

In short, this body of work constructed a picture of how young children's undifferentiated ability/effort concept functions in ordinary achievement situations. If young children are unable to distinguish between ability and effort, as the canonical account proposed, then one would not expect them to be demotivated by clear evidence of their inadequacy. Further, this picture of young children's optimism and resilience stood in sharp contrast to the image of older children and adults, who usually adjust evaluations of their own abilities following a failure experience or after learning that peers performed better than they did (e.g., [Ruble et al., 1980](#)). It seemed quite reasonable then, that the concepts underpinning these behaviors—namely, ability and effort—might undergo qualitative change and restructuring with age.

1.1 Problems for the canonical account of children's ability concepts

There was, at the same time, some evidence to suggest that this theory did not provide an adequate account of the development of children's ability concepts. It did not fit well with contemporaneous findings showing that in some cases, fairly young children could in fact make logical inferences about varying levels of ability, effort, and outcomes. In one study, 6-year-old children judged that a less-able student would have to try harder than a more-able student to complete a puzzle ([Karabenick & Heller, 1976](#)). Similarly, even younger children, 4-year-olds, seemed to understand that a character with high ability who obtained a poor outcome exerted little effort, and that a character who exerted little effort but obtained a good outcome had high ability ([Wimmer et al., 1982](#)).

There were also reasons to believe that the rosy canonical view of young children's reactions to failure was not entirely accurate either. Even quite young children—4- to 6-year-olds—seemed to show individual differences in how they responded to challenges and setbacks ([Burhans & Dweck, 1995](#); [Heyman et al., 1992](#); [Smiley & Dweck, 1994](#); see also [Butler, 2005](#); [Kagan, 1981](#)). Much like older children and adults, many young children displayed decreased motivation, including negative affective responses and lower persistence, after learning that they were unsuccessful in completing a task ([Heyman et al., 1992](#)). The canonical account also did not fit well with findings showing that children's behavior does, under some circumstances,

seem to be responsive to ability-relevant information (Cimpian et al., 2007; Zhao et al., 2017, 2018; see also Rhodes & Brickman, 2008). In one study, for example, 4-year-old children's responses to failure were more negative when the failure was framed as resulting from lack of ability rather than lack of effort (Cimpian et al., 2007). Failure and setbacks could, for many young children, be demotivating experiences. So then why did young children seem to have a single, undifferentiated concept mixing ability and effort in some cases, but not others? Were their ability concepts truly undifferentiated, or could task demands have prevented them from accessing and deploying a differentiated concept of ability?

Today, there is considerable evidence that undermines the claim that young children have an undifferentiated ability/effort concept. Consider, for example, that by age 5 children distinguish ability from performance in surprisingly sophisticated ways. They judge that a more able runner who loses a race to a less able runner because of an intervening event (e.g., tripping over a stone) would, in the future, perform better than the less able runner (Yang & Frye, 2016). This suggests that young children have some notion of ability as a personal resource that—barring unusual intervening events—is consistent in its effects on performance across situations.

Children's understanding of ability is further revealed in their reasoning about how ability relates to others' mental states. When two puppets complete a low-effort task such as climbing a short structure to obtain a reward, but one of the puppets has previously expressed a preference for a reward requiring high effort (i.e., climbing a tall box), 5- and 6-year-old children infer that the puppet with the preference cannot climb (Jara-Ettinger et al., 2015). Further, even younger children understand that low (vs. high) ability corresponds with different mental states about, or subjective evaluations of, tasks: Three-year-olds understand that a person who found a task hard has less ability than a person who found the task easy (Heyman et al., 2003).

In these cases, it is clear that children's concepts of ability have some degree of nuance and sophistication. However, more direct evidence of young children decoupling ability from *effort* is really needed before concluding that they do in fact have a differentiated ability concept. To that end, recent work has reconsidered the most important empirical evidence for the canonical account of children's ability concepts by examining features of the two-student paradigm discussed earlier (e.g., Nicholls, 1978). Two features stand out. First, in these older paradigms the student who works less hard is always depicted with potential misbehaviors, such as playing with a textbook or pencil or looking around the room. Not only was this conduct less

than ideal, but it was also in the context of the *classroom*. This specific presentation of low effort could have unduly influenced the responses of young children, who are greatly concerned with (and inundated with messages about the importance of) behavioral conduct and work habits (e.g., Stipek & Tannatt, 1984). This may have led children to view the student who works less hard—and their ability—more negatively than they would have in the absence of such behaviors. In addition to their being (viewed as) impermissible, these behaviors may not have been a good signal of a *lack of a need* to work hard—instead, they could have signaled, and been interpreted as, trouble with the schoolwork or disliking of it.

Second, after children viewed the two-student paradigm, they were always *first* asked by the experimenter if one of the students worked harder or if they worked the same (Nicholls, 1978; Nicholls & Miller, 1984b). It was always after children had responded to this question that they were then asked to evaluate the students' abilities. This is important because the question about effort may have incidentally set up a task demand that influenced subsequent evaluations of the students. After explicitly stating that the hardworking student put more effort into their schoolwork, young children may have had a globally positive view of this student, and thus assessed their abilities more favorably afterward. Overall, then, young children may have been anchored by *task demands* (here, the behaviors of the non-hardworking student and the order of the questions), such that when they later responded to questions about the students' respective abilities, they gave more positive evaluations about the hardworking student.

So perhaps young children do understand ability but had difficulty deploying these concepts in the presence of these two features. Maybe the question young children were answering was closer to “Who has more ability: Someone who performed well and behaved? Or someone who performed well and didn't behave?” In fact, later studies that disambiguated these two features by (a) including a clearer signal of a lack of a need to work hard (earlier completion of the schoolwork) and (b) not asking the question about effort first found that even very young children say that the less-hardworking student is smarter than the hardworking student (Heyman & Compton, 2006; Muradoglu & Cimpian, 2020). For example, in one study, 4- to 9-year-old children were shown vignettes of two students working side-by-side on schoolwork (Muradoglu & Cimpian, 2020). In one condition, children were shown the non-hardworking student working *intermittently* on their work, as in the canonical paradigm, and in another condition, children were shown the non-hardworking student working

continuously (and finishing the task *before* engaging in off-task behavior). The latter condition, where low effort was depicted as continuous rather than intermittent, was designed to be a clearer signal of a lack of a need to work hard. Importantly, however, the amount of effort put in by this student, as operationalized by number of frames in a storybook, was equivalent across the two conditions. What varied was simply whether their effort was depicted as intermittent throughout the task or concentrated and continuous at the beginning of the task.

In this study, after children were shown the vignettes, they were then asked to assess each student's effort and ability. Children in one condition were asked the effort question immediately after viewing the vignettes, as in the canonical paradigm. Children in another condition were asked to make judgments about how hard the students found the schoolwork after viewing the vignettes. Here again, the latter condition was designed to prevent children from being unduly anchored by positive assessments of the hardworking student.

When children made judgments about the students' abilities *without* the two features of the canonical paradigm, they rated the less-hardworking student's abilities more highly than the hardworking student's, suggesting that ability and effort are differentiated by young children. Even the youngest children—4-year-olds—distinguished the two. Importantly, children still rated the non-hardworking student as working *less* hard than the hardworking student, suggesting that children can indeed successfully decouple ability from effort (Muradoglu & Cimpian, 2020; see also Leonard et al., 2019, for additional evidence that young children consider speed and performance information together). And, there was support for the idea that the intermittent display of effort on behalf of the less-hardworking student signaled difficulty with the task. Children's perceived difficulty ratings for the less-hardworking student were higher in the intermittent-low-effort condition than in the continuous-low-effort condition, and this shift in perceived difficulty mediated the effect of type of low effort on children's ability judgments.

These findings offer more compelling evidence against the claim that children do not acquire a mature, differentiated concept of ability until around age 10 by showing that much younger children—4- and 5-year-olds—have concepts of ability and effort that are differentiated (see also Cimpian, 2017). What appeared to be conceptual immaturity may have instead been the result of methodological features that masked children's conceptual sophistication.

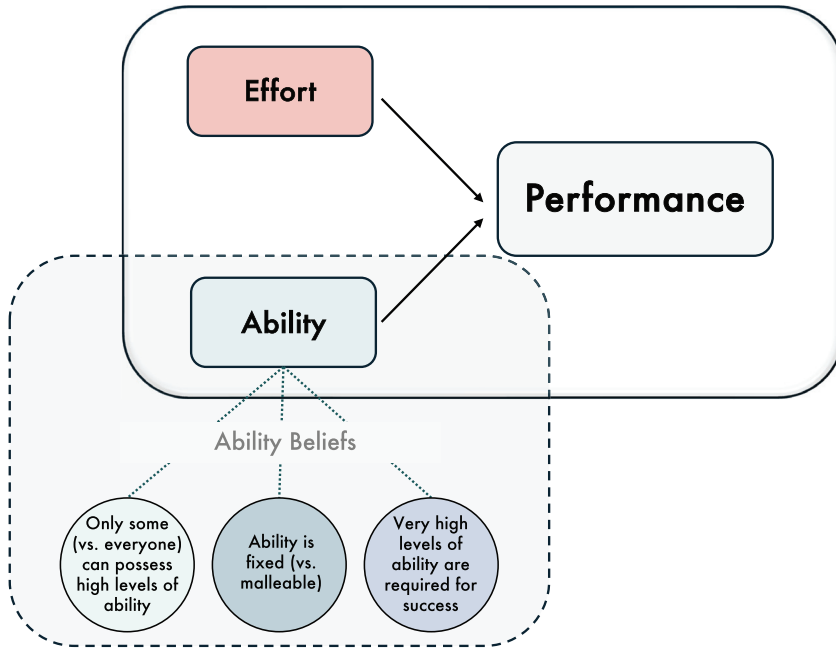


Fig. 2 Schematic of ability as a concept, distinct from other performance inputs, such as effort. Ability, like effort, bears on performance outcomes, and people can hold a number of nuanced beliefs about the nature of ability (in blue circles).

2. Beyond having a differentiated ability concept: The organization and impact of children's beliefs about the nature of ability

The aim of the prior section was to address whether young children have a unitary concept of ability and effort that increases in differentiation and clarity with age. The weight of the evidence suggests that this is not the case. Even very young children seem to initially have intuitive theories in which ability and effort are distinct concepts. However, it is one thing to have a notion of ability as a distinct cause, and yet another to have assumptions about what ability is like—that is, to have beliefs about its nature (see Fig. 2). These assumptions or beliefs involve content about what is true (and untrue) about ability. If, as we claim, young children do in fact have a clear concept of ability, then they might also build beliefs around this concept concerning what ability is like. And, if children have beliefs about what ability is like, these beliefs might be *motivationally relevant*—that is, they may be involved in

Table 1 Beliefs about ability, their conceptual meaning, and references.

Belief	Conceptual definition	References
Growth (vs. fixed) mindsets	Beliefs about the malleability of ability	Dweck, 1999; Dweck and Leggett, 1988; Dweck et al., 1995
Universal (vs. nonuniversal) mindsets	Beliefs about the distribution of potential for high ability in a population	Rattan et al., 2012; Rattan et al., 2018; Savani et al., 2017
Brilliance beliefs	Beliefs about the extent to which brilliance—that is, a very high level of ability—is required for success	Jenifer et al., 2024; Leslie, Cimpian et al., 2015

children’s learning behaviors and achievement. The aim of this next section is to examine these possibilities.

We begin by summarizing theory and research on a set of beliefs with a long history in the field of achievement motivation: beliefs about how *malleable* ability is (Cain & Dweck, 1995; Dweck & Bempechat, 1983; Dweck & Leggett, 1988). This work suggests that such beliefs—now called “growth (vs. fixed) mindsets”—are foundational to motivation, learning, and achievement (Blackwell et al., 2007; Dweck & Leggett, 1988; Hong et al., 1999; Limeri et al., 2023; Rege et al., 2021; Yeager et al., 2019).

Yet, growth (vs. fixed) mindsets are only one part of a dense network of beliefs about ability’s properties. As adults, our concept of ability is richly detailed with assumptions concerning not only how *malleable* (vs. static) it is, but also whether high levels of it are required for success in specific contexts, whether everyone (or just some people) can possess the highest levels of it, and so on (see Fig. 2). Over the years, researchers have expanded empirical inquiries beyond growth (vs. fixed) mindsets to consider these beliefs—called “brilliance beliefs” and “universal (vs. nonuniversal) mindsets,” respectively (see Table 1).²

Are young children’s ability concepts richly detailed with these kinds of ideas? Or might children’s ideas about ability be more limited than those of adults? Further, if children *do* have a rich set of assumptions about the nature of ability, do these assumptions link up with each other in logically coherent ways, or are they somewhat scattered and independent? Here again

² Importantly, this list of beliefs—growth mindsets, universal mindsets, and brilliance beliefs—is not an exhaustive one.

it makes sense to first review the state of the evidence from investigations with older children (i.e., of middle- and high-school age) and adults before considering evidence from young children, which is relatively more limited. In what follows, we first describe these nuanced beliefs in more detail and then present evidence on their structure—that is, their dimensionality and interrelations—among adults. We then turn to emerging evidence that points to the presence of a structured, coherent, and motivationally active network of ability beliefs in early childhood.

2.1 Components in the network of beliefs about ability

2.1.1 *Growth (vs. fixed) mindsets*

A long tradition of scholarship has looked to a particular type of belief to understand the dynamics of motivation: beliefs about how *malleable* ability is (Dweck, 1986; Dweck & Leggett, 1988). Some people view ability as relatively dynamic and malleable (i.e., a growth mindset) and others view it as relatively stable and fixed (i.e., a fixed mindset; Cain & Dweck, 1995; Dweck & Bempechat, 1983; Dweck & Leggett, 1988). Assumptions about the malleability of ability have received a great deal of empirical attention because people's beliefs along this dimension have been shown to influence learning behaviors, and, as a result, achievement (Blackwell et al., 2007; Claro et al., 2016; Dweck, 1986; Dweck & Leggett, 1988; Dweck et al., 1995; Hong et al., 1999; OECD, 2019; Yeager et al., 2019). These links are present because ability beliefs can influence and structure key motivational processes, including people's goals (i.e., to increase mastery or showcase ability), understandings of effort (i.e., as useful or futile; as a way to increase ability or a sign of low ability), and attributions for and responses to failure (e.g., increase or withdraw effort) (Blackwell et al., 2007; Hong et al., 1999; Robins & Pals, 2002).

For instance, at the level of goals, people with growth mindsets tend to be more focused on increasing mastery of skill, in comparison with those with fixed mindsets, who are more focused on *showcasing* their abilities and preventing anything that might discredit them (e.g., Blackwell et al., 2007; Dweck & Leggett, 1988). Because beliefs about the malleability of ability predispose people to more (vs. less) adaptive approaches to and responses to achievement events, these beliefs can end up contributing to actual achievement outcomes as well. There is some evidence to suggest that, compared to people with stronger fixed views of ability, people with stronger growth views of ability tend to earn higher grades in school

(Blackwell et al., 2007; Romero et al., 2014; Yeager et al., 2016) and to perform better on standardized tests (Claro et al., 2016; Cury et al., 2008; McCutchen et al., 2016).

2.1.2 Universal (vs. nonuniversal) mindsets

One extension of work on growth (vs. fixed) mindsets concerns universal (vs. nonuniversal) mindsets. Whereas growth and fixed mindset beliefs provide answers to the question “*Can ability be developed?*” universal and nonuniversal beliefs provide answers to “*Can anyone (or only some people) possess the highest levels of ability?*” (Rattan et al., 2012, 2018; Savani et al., 2017). Universal beliefs involve assumptions that virtually everyone has the capacity to possess the highest levels of ability; nonuniversal beliefs involve assumptions that only some have the capacity to reach these levels of ability. To make the distinction with growth (vs. fixed) mindsets more concrete, consider the person who believes that ability can be increased (i.e., has a growth mindset), but simultaneously believes that the *highest* levels of ability can only be reached by a select few (i.e., has a nonuniversal mindset).

Prior work shows that these beliefs among instructors can play a role in students' motivational processes. In particular, students' *perceptions* of the extent to which universal (vs. nonuniversal) beliefs are endorsed by others in their learning environments seem to be important for students' sense of belonging. When students perceive that their instructors believe that *everyone* has the potential to possess the highest levels of ability (i.e., hold a universal mindset), students feel a stronger sense of belonging to their field. Further, students who had these perceptions of their instructors' beliefs outperformed students who perceived their instructors to hold nonuniversal mindsets, a difference that was explained in part by these differences in belonging (Rattan et al., 2018).

2.1.3 Brilliance beliefs

Another dimension that has been recently prominent in the study of ability beliefs involves the *importance of very high ability for success* in certain arenas. Some view very high levels of intellectual ability (i.e., “brilliance”) to be a precondition for success in a particular domain or context, while others view this level of ability as less necessary for success, with effort and motivation being sufficient (Leslie, Cimpian et al., 2015; Muradoglu et al., 2023, 2025). Whereas growth (vs. fixed) mindsets and universal (vs. nonuniversal) mindsets concern the perceived *properties* of ability (that is, its malleability and its distribution in the population, respectively), brilliance beliefs instead

concern ideas about ability's *importance*. These beliefs are often brought to bear when considering what it takes to excel in specific domains or areas of study. For instance, people tend to think that brilliance is necessary for success in domains such as mathematics, physics, and philosophy (Ito & McPherson, 2018; Leslie, Cimpian, et al., 2015; Rutten et al., 2024).

Importantly, endorsing such beliefs about brilliance is thought to thwart processes involved in motivation, particularly for some groups of people. In one study, for example, high school girls who believed brilliance to be central to success in the physical sciences, technology, engineering, and mathematics expressed weaker intentions to pursue such domains (Ito & McPherson, 2018). Similarly, environments whose members valued brilliance for success tended to prompt lower belonging and higher anxiety (Bian et al., 2018), as well as impostor feelings (Muradoglu et al., 2022), particularly among members of underrepresented groups.

2.2 The dimensionality, coherence, and correlates of ability beliefs among adults

Though conceptually distinct, these beliefs bear enough resemblance to each other that one might expect some amount of interdependence among them. For example, one can reasonably expect people with growth mindsets to more often hold universal (vs. nonuniversal) beliefs about ability, and to view brilliance as less important for success. Similarly, one might expect people with fixed mindsets to more often hold nonuniversal (vs. universal) beliefs about ability, and to view brilliance as more important for success. Empirically speaking, to what extent are these beliefs interconnected? And, if interconnected, *how* interconnected are they? Might they be so highly related that they are best understood as representing the same underlying construct?

The empirical evidence so far does not support the idea that these beliefs reflect a single underlying construct. Instead, growth (vs. fixed) mindset beliefs, universal (vs. nonuniversal) mindset beliefs, and brilliance beliefs seem to be distinguishable but related dimensions. First, correlations between growth (vs. fixed) mindsets and universal (vs. nonuniversal) mindsets range from .14 to .27, suggesting distinguishability but some interdependence (Rattan et al., 2012). Second, a more recent study that examined growth mindset beliefs, universal mindset beliefs, and brilliance beliefs together in college students found that these beliefs were empirically represented by separate underlying dimensions (Limeri et al., 2023). Though distinct,

these beliefs showed a logical pattern of interdependence: Students who believed that ability was malleable tended to also believe that anyone could reach the highest levels of it ($r = .38$), and that brilliance was not a precondition for success (in STEM) ($r = -.24$). Students who believed that anyone could reach the highest levels of ability tended to also believe that brilliance was not a precondition for success (in STEM) ($r = -.42$) (Limeri et al., 2023; see also Bian et al., 2018; Porter & Cimpian, 2023). Thus, adults seem to carry with them an internally consistent belief system about the nature of ability. Importantly, however, the magnitude of the correlations between these beliefs point to interdependence but distinguishability.

This belief system also appeared to be bound up with motivational processes among students. Consistent with theory (Dweck & Leggett, 1988), students' growth mindsets were associated with their goals, such that students who viewed ability to be relatively malleable had goals for mastering course material. Also consistent with theory, these students were relatively less focused on simply avoiding performing worse than their peers. Further, compared to students who thought that ability was relatively unmalleable, students who thought that ability was relatively malleable had lower levels of evaluative concern—that is, they were less anxious about the prospect of their abilities being judged negatively by others. Students who viewed ability as relatively universal, as well as students who thought that success (in STEM) was not strongly dependent on brilliance reported weaker levels of evaluative concern as well (Limeri et al., 2023).

2.3 Evidence for distinct, coherent, and motivationally active ability beliefs in early childhood

An important developmental question concerns the extent to which these dimensions can be reasonably described as a coherent, motivationally active network of beliefs among children. This broad question can be broken down into three more concrete ones. First, are children's ideas about ability more limited than those of adults? Second, if children do have a rich set of assumptions about the nature of ability, do these assumptions link up with each other in logically coherent ways, or are they somewhat scattered and independent? Third, to the extent that these beliefs are distinguishable, are they motivationally *active* among young children? In other words, are key learning behaviors, such as embracing challenges, tied to children's assumptions about ability?

There is some reason to expect that children's ability beliefs might, like adults', be multifaceted and coherently interrelated. Indeed, at around age 5, children start to acquire culturally constructed ideas about ability, and their ideas show a notable degree of consensus with those of adults. For example, as early as first grade, children view high levels of intellectual ability to be more central to success in domains such as math than in domains such as reading and writing (Gunderson et al., 2017; Jenifer et al., 2024; see also Muradoglu et al., 2025). Also, at around age 6, children begin to associate high levels of intellectual ability with boys and men more than with girls and women (Bian et al., 2017; Jaxon et al., 2019; Kim et al., 2024; Okanda et al., 2022; Shu et al., 2022). In addition to ideas about who is *generally* more intellectually talented, young children seem to have adult-like stereotypical beliefs about who is more intellectually talented in certain domains. On both explicit and implicit measures, second-grade children associate mathematics with boys more than with girls (Cvencek et al., 2011; Miller et al., 2024). Similarly, around age 5, children begin to view men as more competent than women in science, technology, engineering, and mathematics (STEM; Shenouda et al., 2024).

Although these findings hint at detailed, adult-like beliefs, they do not speak to the organization or coherence of children's beliefs about the *properties* of ability. One recent study, however, offered more direct evidence on this matter by embracing the full nuance of these beliefs (Muradoglu et al., 2025). In this study, the authors examined 5- to 11-year-old children's beliefs about ability's malleability, its universality, and its importance for success.³ To ensure that items tapping these beliefs were accessible for young children, the language concerned abilities in math and spelling (e.g., "good at math"), which represent concrete manifestations of ability, rather than general intellectual ability.

The authors were first interested in understanding the *structure* of these beliefs, asking whether they represent truly distinct dimensions of reasoning about ability (as seems to be the case among adults) or instead form a single belief amalgam. Though young children clearly seem to have some adult-like understandings of ability, there are still grounds for expecting their beliefs about its properties to be not neatly distinguishable. Consider, for instance, work on the development of self-concepts that has shown that young children can make self-assessments regarding their competencies

³ The authors measured two other beliefs (innateness and responsiveness to intervention) for a total of five; however, for brevity and clarity, we do not discuss these additional two beliefs here.

in cognitive and physical domains (Harter, 1982; Harter & Pike, 1984). However, for children younger than 8, the structure of their self-assessments seemed to lack dimensionality. Rather than loading on two distinct factors, young children's self-assessments in cognitive and physical domains loaded onto a *single* factor. This was interpreted as evidence that the self-structure becomes more differentiated and dimensional with age (but see Marsh et al., 2002, for evidence that 4-year-olds distinguish between multiple dimensions of self-concept).

What were the dimensions underlying young children's ability beliefs? The results indicated that items tapping the beliefs of interest—ability's malleability, universality, and importance for success—loaded onto three separate factors. In addition, this structure appeared to be consistent both among the younger children in the sample (before about 8 years of age) and among the older children (Muradoglu et al., 2025). This suggests that although the three beliefs share a “family resemblance,” even young children's mental representations differentiate between them. Because the structure of children's beliefs about ability does in fact reflect these nuances, it does not seem sensible to treat these dimensions as missing or merged.

The second main question addressed in this study concerned coherence: To what extent are these three beliefs correlated with one another? The results indicated that these beliefs were linked in a logically coherent manner: Children who thought that intellectual ability was relatively fixed also tended to endorse the idea that it was nonuniversal and that high levels of it were a precondition for success. Similarly, children who viewed ability to be relatively malleable tended to also view it as relatively universal, and they thought that high levels of it were not a strong requirement for success. These (disattenuated) correlations ranged from .32 to .64 in absolute value, suggesting small to moderate levels of concordance—and, importantly, non-redundancy (Muradoglu et al., 2025).

The third and final question in this study concerned the motivational significance of these beliefs: Are children's ability beliefs implicated in their motivation? Two findings suggested the answer is “yes” (Muradoglu et al., 2025). First, above and beyond age, children who viewed ability to be malleable (rather than fixed) opted for learning goals (rather than performance goals) and preferred more challenging tasks. Second, younger children who thought that very high levels of ability (i.e., brilliance) were a precondition for success in school reported higher levels of evaluative concern—that is, they were more sensitive about how their abilities might be perceived by others in academic contexts (see Fig. 3). These

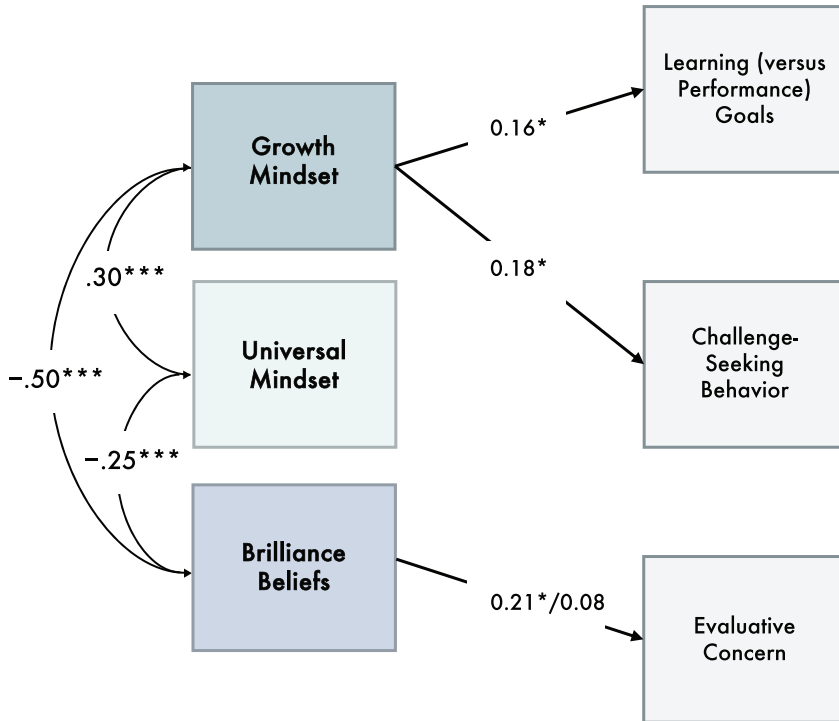


Fig. 3 Path model illustrating the relations among growth mindset, universal mindset, and brilliance beliefs, as well as relations between beliefs (growth mindset, universal mindset, and brilliance beliefs) and motivation (goals, challenge-seeking, and evaluative concern). Numerical entries are estimates of correlated residual variances (zero-order correlations) and standardized regression coefficients, respectively. Child age moderated the association between brilliance beliefs and evaluative concern; therefore, the path shows regression coefficients for younger and older children separately (i.e., younger/older). All other paths from beliefs to motivation adjust for child age. The data come from [Muradoglu et al. \(2025\)](#). * $p < .05$. ** $p < .01$. *** $p < .001$.

relationships are both intuitive and aligned with theory ([Dweck & Leggett, 1988](#); [Dweck et al., 1995](#)): An understanding of ability as a relatively fixed and highly important resource sets up a “world” in which the priority is to “prove and protect” one’s abilities. It makes sense that in this “world,” one would gravitate toward easier, lower-risk tasks and experience concerns about how others might perceive one’s abilities.

Together, these findings suggest that even young children have a network of coherently organized beliefs about ability. Children’s beliefs about ability’s malleability, its universality, and its importance for success appear to be clearly distinct dimensions. Further, these beliefs are coherent, in that most of them

correlated with one another in logically expected ways. Children with relatively stronger (vs. weaker) beliefs about ability's malleability also had relatively stronger (vs. weaker) beliefs about ability's universality, as well as relatively weaker (vs. stronger) beliefs about ability's importance for success. A final piece of evidence suggests that in addition to being distinct and coherently interrelated, these beliefs are motivationally active: Children's motivation appears to be, at least in part, organized around their beliefs about what ability is like. Children's beliefs about ability's malleability and importance for success were linked with the extent to which they embraced challenges and adopted goals for increasing their mastery, and with the extent to which they expressed concern over others negatively judging their abilities, respectively. Although it is not possible to draw causal conclusions here, the results nevertheless suggest that even quite young children might, much like adults, enter achievement situations with a network of ability beliefs that has some degree of sophistication, coherence, and motivational force.



3. Conclusions and avenues for future research

The literatures we have reviewed here yield three key conclusions about the psychology of ability in early childhood. First, children seem have a well-formed concept of ability, distinct from effort, by age 4 or 5. Second, by the age of 5, children also appear to have a multifaceted and coherently interrelated set of beliefs about what ability is like. Third, this set of beliefs is not only coherent but also already linked up with children's motivation. Before describing avenues for future research based on these conclusions, it is useful to briefly consider the impact of the older, "classic" arguments on the field's views of, and approach to studying, young children.

Because young children were painted as relatively unconcerned with ability-relevant information, they were regarded as both unshakably optimistic and highly resilient to failure (but see [Heyman et al., 1992](#); [Smiley & Dweck, 1994](#)). This picture predictably led to efforts to understand (and ultimately improve) motivational processes among older age groups, which were assumed to be more vulnerable: namely, older children, adolescents, and adults. It is no wonder that our understanding of basic motivational concepts and processes in early and middle childhood is relatively limited as a result. And, admittedly, what we do know now may still be a somewhat incomplete account of young children's motivational concerns and problems. Our hope is that the revised picture we have proposed here, along with our suggestions

for future research, will contribute to a deeper and more complete account of motivation in early childhood.

3.1 Avenues for future research

A key upshot of the work we have summarized here is that young children have sophisticated, motivationally active concepts of ability much earlier in development than previously theorized. One way of thinking about this is that we now have a much clearer picture of “the seeds”—that is, of the *beliefs* and *concerns*—that guide children’s motivation. Yet these beliefs and concerns do not develop in a vacuum. A useful (and we would argue, valid) way of thinking about children’s ability beliefs, and their motivational tendencies more generally, is to regard them as deeply embedded in (social) contexts (de Ruiter & Thomaes, 2023; Haimovitz & Dweck, 2017). A generative avenue for future research, then, will be to better understand these contexts—that is, to better understand “the soil” in which “the seeds” take root and grow (Walton & Yeager, 2020).

In the pursuit of better understanding children’s contexts, one more general priority may be to make progress toward building a rich understanding of the ordinary, everyday moments that shape the beliefs and motivational tendencies children come to adopt. Our current understandings of young children’s quotidian exposure to ability- and achievement-relevant cues, for example from teachers or parents, is somewhat limited. For instance, when a child makes a mistake in their schoolwork, how (if at all) do teachers respond? When children receive explicit evaluations of themselves in the form of tests or report cards, what do conversations between teachers (or parents) and children look like? When children are placed in ability groups, what are they told, if anything? We believe that research on children’s everyday achievement-related experiences, including some of the events we have outlined here, could contribute substantially to an understanding of how academic contexts shape early motivational processes. In addition to its theoretical contribution, a clearer picture of the contextual factors that shape young children’s beliefs about ability—that is, “the soil”—would lay the groundwork for intervention research to improve young children’s beliefs and motivational tendencies.

3.1.1 Home contexts

Although much remains to be studied about these experiences, a few literatures offer useful starting points. One is work on feedback from adults. A

branch of this work has focused on how so-called process praise (e.g., praise for one's effort or strategy) versus person praise (e.g., praise for one's ability) might shape the mindsets children adopt and their vulnerability to setbacks. This research has taken the form of carefully controlled experimental studies (Kamins & Dweck, 1999; Mueller & Dweck, 1998) as well as field studies whose focus was on praise from parents (Barger et al., 2022; Gunderson et al., 2013, 2018; Pomerantz & Kempner, 2013). The field studies were with children of very different ages and used different methods and measures. They, not surprisingly, yielded heterogeneous patterns of results, with some studies finding links between process (but not person) praise and children's mindsets (Gunderson et al., 2013, 2018), some finding links between person (but not process) praise and children's mindsets (Pomerantz & Kempner, 2013), and others finding no links between either type of praise and children's mindsets (Barger et al., 2022).

Rather than regarding this heterogeneity as a weakness, it can be viewed as a starting point and an opportunity for further investigation. Because field studies involve capturing processes in highly complex and multifaceted environments, perhaps one reason for their heterogeneous results is that some parents express process-oriented praise but enact other behaviors that contradict or undermine the message that it is effort or strategy that matters (see Walton & Yeager, 2020; Wang et al., 2024). In addition, children of different ages might be more or less sensitive to the different kinds of feedback. Finally, in some cases, the parent feedback might be overshadowed by strong feedback from teachers. Future research can (a) seek to resolve these inconsistencies by focusing on when and for whom adult feedback influences children's beliefs, and (b) try to broaden the scope of inquiry on this question to consider antecedents other than praise (e.g., Haimovitz & Dweck, 2016; Zarrinabadi et al., 2021).

3.1.2 School contexts

Any account of how contexts shape children's thinking about ability would be incomplete without attending to one of children's most important ecological "niches": school. Empirical attention to children's educational environments is all the more pressing in light of more general, secular trends toward increasingly ability-focused contexts. Over time, children with different academic abilities have become less likely to attend the same school (OECD, 2015). And, among 80 countries, the US ranks fifth in having one of the highest percentages of students attending schools that sort students by ability (OECD, 2023). Evaluation is very clearly a feature of every

educational context, and recent evidence suggests that children as young as 3 care about how their abilities are evaluated by others (Asaba & Gweon, 2022).

Yet we know very little about the interplay between the features of young children's school environments and their thinking about ability (including their thinking about how their own abilities will be viewed by others) and their motivation. Children's views of ability could well be shaped by teachers' classroom practices and by teachers' expectations of their students (Brummelman & Sedikides, 2023; Darnon et al., 2023; Goudeau & Cimpian, 2021). Therefore, empirical attention to teachers' expectations for (and treatment of) higher- and lower-achieving students, as well as children's perceptions of these dynamics, could lead to valuable insights. Indeed, some researchers are now focusing on these very types of practices in the classroom and coaching teachers in the implementation of practices that shape students' active engagement with learning and pursuit of challenging work (e.g., Hecht et al., 2023a, 2023b).

Another useful endeavor may be to examine the course of ability beliefs and motivational processes across early and middle childhood as a function of the types of school environments children are in. Early theorizing on this topic suggested that as children progress through school, the character of their school environment changes to become more regimented, evaluative, and ability-focused (Eccles et al., 1984). This led many researchers to assume that children came to view ability as more fixed with age.

Although the idea that grade-related changes in school environments shape children's beliefs and motivational orientations is, at face value, compelling, it has not yet, to our knowledge, moved from theorizing to *testing*. Further, the empirical yield on whether and how elementary school-age children's beliefs change with age is quite inconclusive. Some studies have shown increasing growth-oriented views of ability with age (Gunderson et al., 2017; Jenifer et al., 2024; Muradoglu et al., 2024, 2025; Pomerantz & Saxon, 2001; Stipek & Gralinski, 1996), while others have shown the opposite—stronger views that ability is fixed with age (Droege & Stipek, 1993; Heyman & Giles, 2004; Pomerantz & Ruble, 1997; Pomerantz & Saxon, 2001). Still others have found no changes with age (Heyman & Giles, 2004; Kinlaw & Kurtz-Costes, 2007; Muradoglu et al., 2024; Pomerantz & Ruble, 1997).

Part of these inconsistencies may stem from different approaches to measuring and operationalizing children's beliefs about ability (a point we return to later). Indeed, age-related changes in children's ability beliefs seem to depend, at least in part, on the specific question children are asked (e.g.,

Heyman & Giles, 2004; Muradoglu et al., 2024; Pomerantz & Ruble, 1997). Further, some of these investigations involved samples of older elementary-aged children (e.g., Pomerantz & Saxon, 2001), whereas others involved younger elementary-aged children (e.g., Jenifer et al., 2024; Muradoglu et al., 2024), making direct comparisons more challenging. Broadening the age range of focus, as well as the suite of questions in future work may clarify these patterns. Another interesting approach—and one that would speak to the influence of the school environment—would be to use longitudinal designs to compare the beliefs of children attending more- vs. less-academically focused elementary schools (see e.g., Stipek & Daniels, 1988).

3.1.3 Cultural contexts

At a broader level than the *school* context is the *cultural* context. A great deal of what we have discussed—ability concepts, beliefs, motivational tendencies, school contexts—can vary across cultural contexts (Holloway, 1988; Li 2003; Li, 2004; Li & Wang, 2004; Stevenson et al., 1990; Sun et al., 2021; Uttal, 1997). Consider, for instance, that children as young as 4 seem to have conceptions of learning that align with their culture's values. Whereas US American children's conceptions of learning seem to be bound up with ability and strategy use, Chinese children's conceptions appear to be bound up with personal virtues such as diligence and persistence (Li, 2004).

The beliefs of important socializers in children's lives exhibit cross-cultural variability as well. One study showed that although mothers in the US, Japan, and Taiwan all view effort to be a more important factor for children's school performance than ability, this difference is much larger among Taiwanese and Japanese mothers, compared to US American mothers (Stevenson et al., 1990). Similarly, compared to Japanese and Taiwanese mothers, US American mothers were less likely to endorse the idea that any student could be good at math or reading if they worked hard enough (Stevenson et al., 1990). It would be informative to explore potential visible manifestations of these beliefs among young children, with attention to how these manifestations may vary cross-culturally. One other important consideration is that, of the estimated 2 billion children in the world, a minority live in Western societies. It is therefore an exciting and important endeavor to study children's ability concepts, as well as their motivational concerns and problems, more globally.

3.1.4 Attention to measurement

We wish to end by acknowledging that a continuing challenge to these pursuits is a lack of concerted empirical attention to a key part of any

psychological investigation: careful measurement of constructs (e.g., Hodson, 2021). One real problem is that measurement is a challenging undertaking in itself, and almost all of the constructs we have focused on here involve content that can be highly abstract. These challenges are only compounded when the target participants are young children, whose still-developing language skills may make it difficult to express their beliefs and thoughts about concepts such as ability. But this does not mean that children do not have abstract thoughts, nor does it mean that it is impossible to access and study such thoughts via careful and serious measurement pursuits (see Barroso et al., 2023; Muradoglu et al., 2024). Though there are likely a number of approaches to increasing comprehensibility, one used in some recent studies (Muradoglu et al., 2024, 2025) has involved using multiple concrete, domain-specific expressions of ability (e.g., “good at math,” “good at spelling”) to approximate children’s beliefs about ability more generally. We look forward to future work that embraces the challenges—and opportunities—inherent in measuring young children’s thinking.

3.2 Conclusion

Ability is a key concept in people’s intuitive theories of performance. We reviewed evidence that calls into question canonical views of children’s ability concepts, showing that core elements of adults’ intuitive understanding of ability are present from early childhood. Our hope is that this effort to provide an up-to-date overview of young children’s ability concepts can serve as a starting point for research that seeks to understand the formation and downstream consequences of young children’s thinking about ability.

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